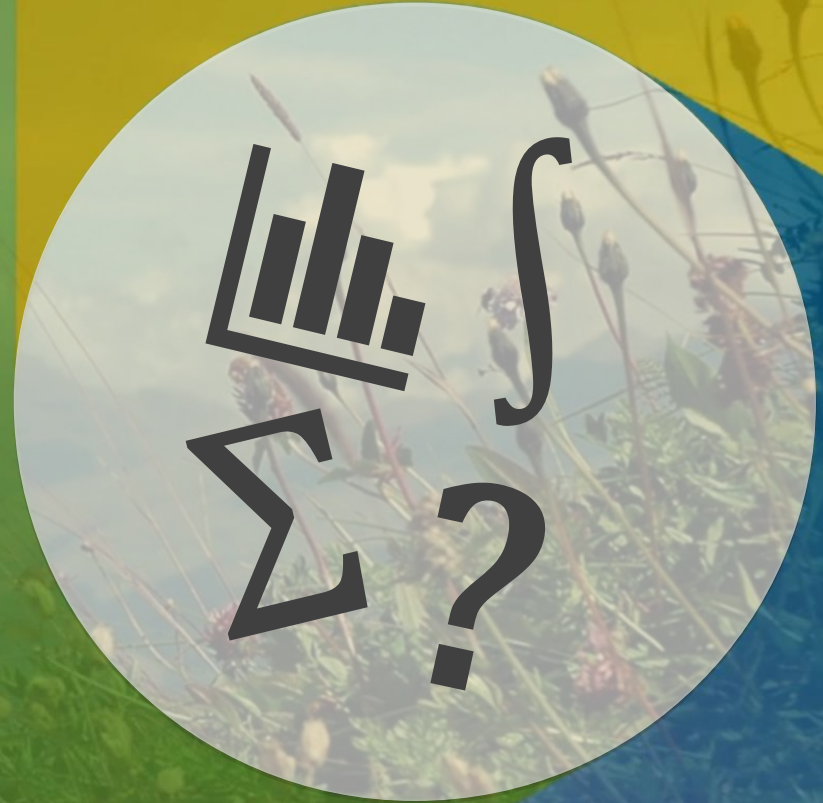


Introduction to Bayesian Statistics

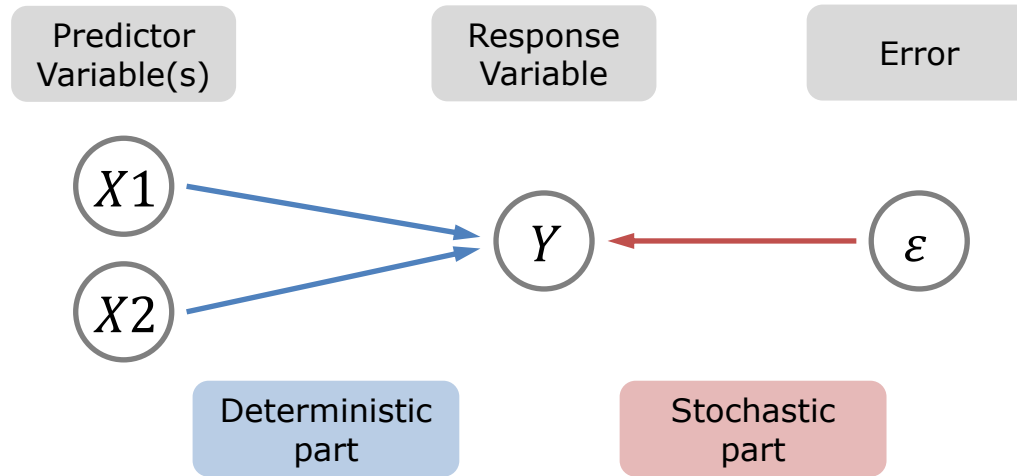
Part 8 Conclusions

Benjamin Rosenbaum



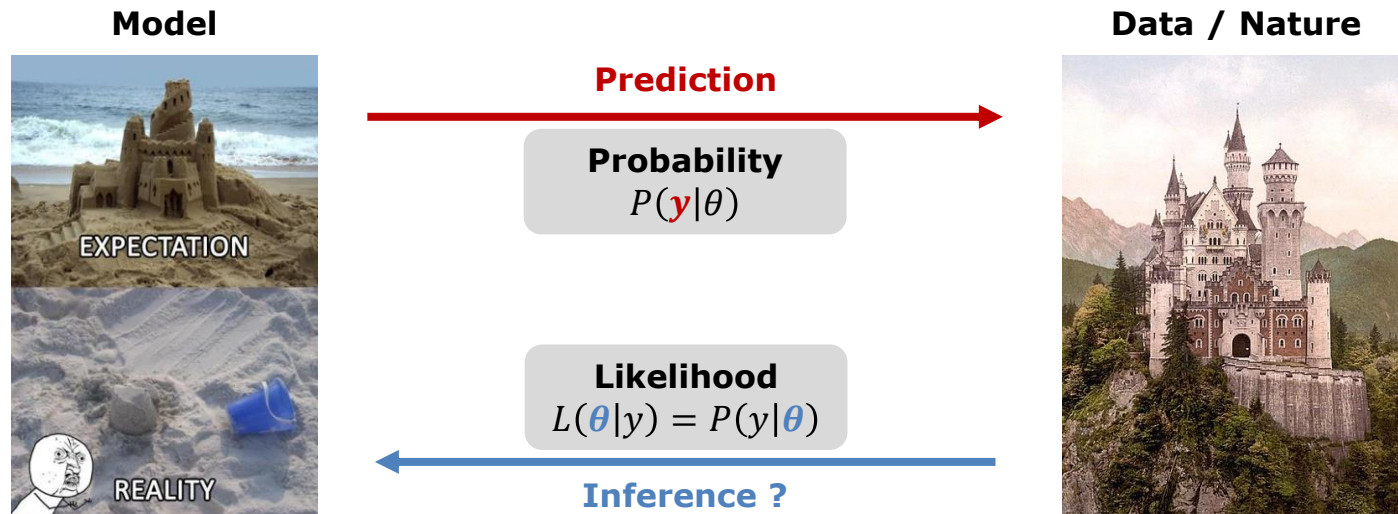
iDiv 2026

Statistical modeling



- Deterministic part:
Prediction model, e.g. mean regression line
- Stochastic part:
The prediction model cannot explain response perfectly, include random error
- Deterministic and stochastic parts both have **parameters** (e.g. effect sizes)

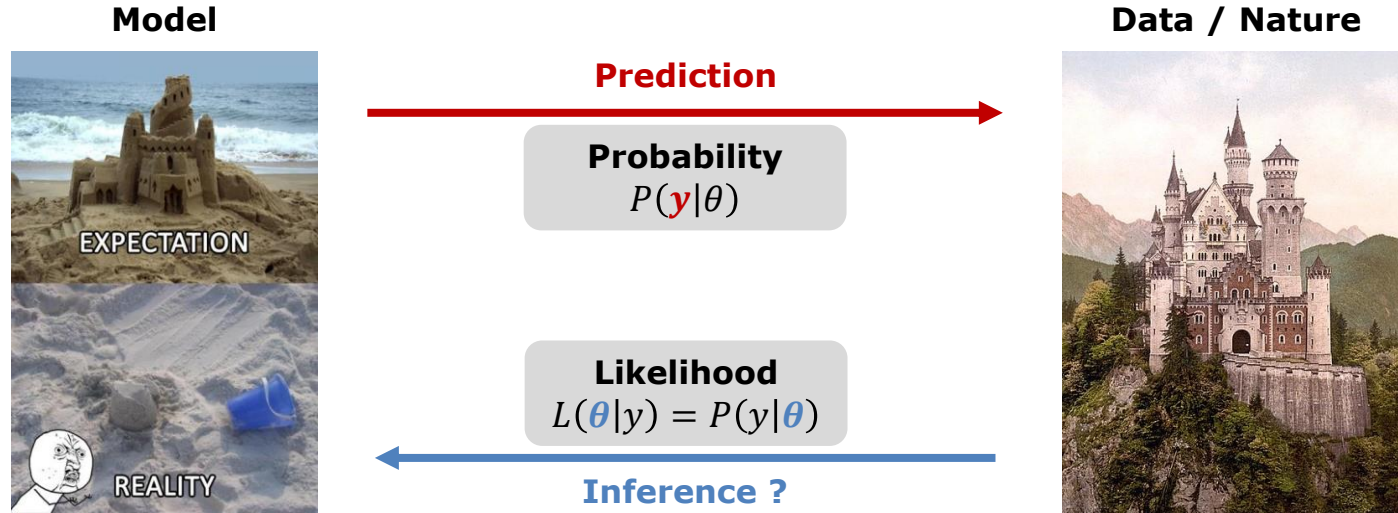
Statistical modeling



We want to make **quantitative statements** on our research questions:

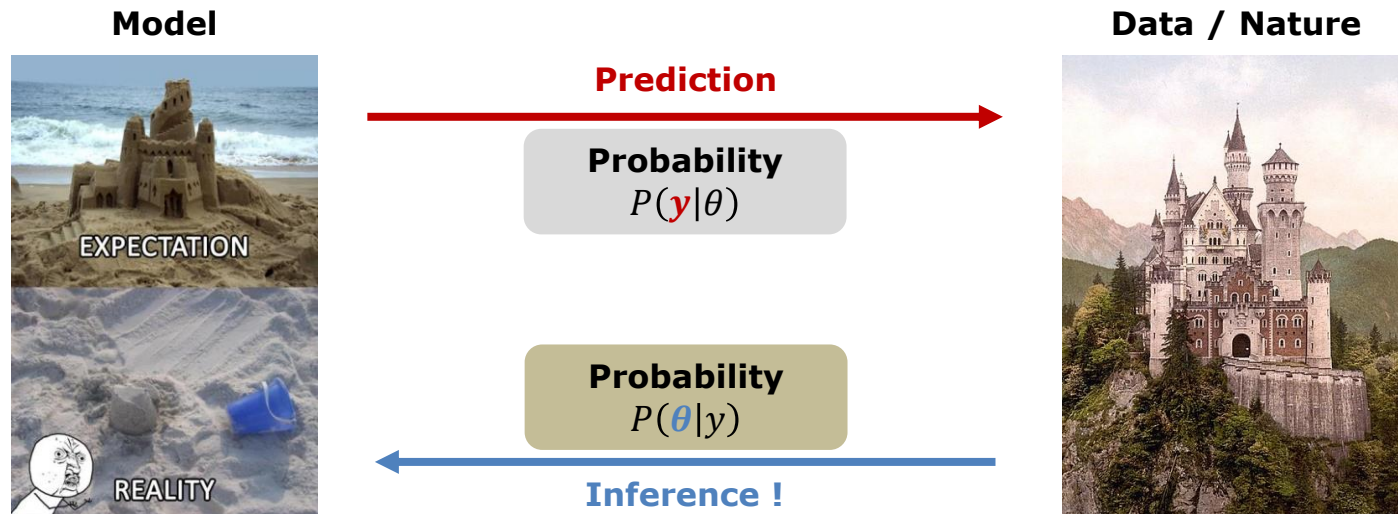
What does the data tell me about my model?

Statistical modeling



Frequentist statistics cannot (mathematically) do direct inference $P(\theta|\mathbf{y})$,
and requires a (methodological) detour via **NHST** $P(\mathbf{y}|\theta = 0)$

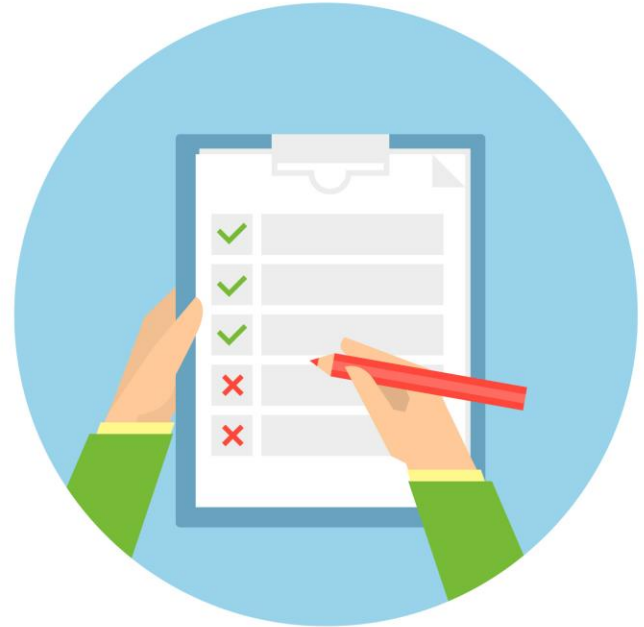
Statistical modeling



Bayesian statistics can (conceptually) do direct inference $P(\theta|y)$,
but requires a (computational) detour via **MCMC**

Bayesian workflow

- 1) Research question (hypotheses)
 - 2) Data collection
 - 3) Statistical model
 - 4) Prior distribution choice
 - 5) Model fitting (MCMC)
 - 6) Evaluate model output
 - 7) Quantitative statements on hypotheses
 - 8) Reporting
- Revise model
- 



designed by  freepik.com

→ Workflow not that different from frequentist statistics.

(1) Research question

(2) Data collection

Your responsibility. You are the experts!

Don't let statistical methods limit your creativity in asking important questions.

But at least think of possible analyses before planning your study / experiments.

The Bayesian 3d printer is more flexible than the frequentist toolbox. But still limited to the framework of statistical modeling, and model identifiability



iDiv Ecotron

Source: Schmidt et al. (2021) Ecol. Evol.
<https://doi.org/10.1002/ece3.8198>

(3) Statistical model

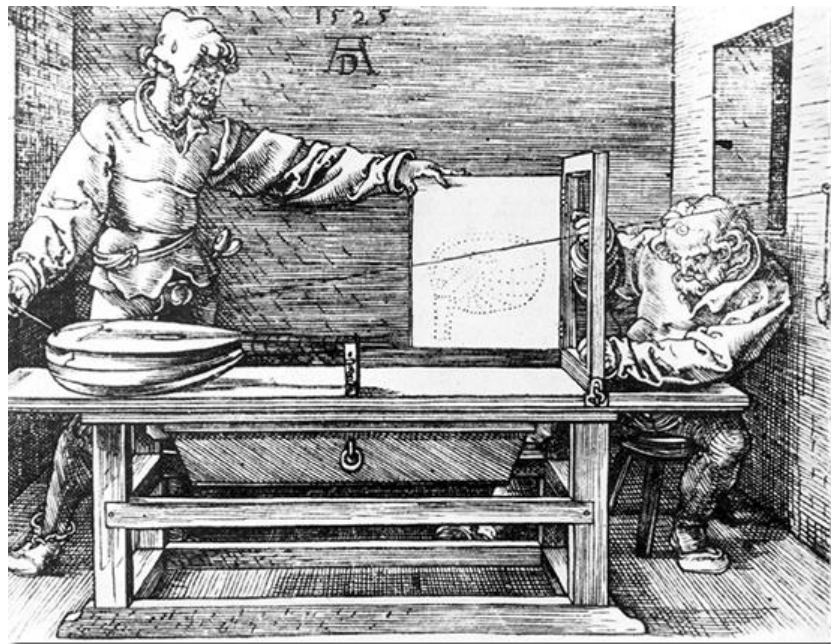
- Driven by research question / hypotheses!

Deterministic part (mean fitted response)

- Start small. Especially when using Stan.
 - few predictors
 - without interactions
 - simple or no random effects structure
- Not as „forward model selection“ ...
- ... but to get an idea how the model works

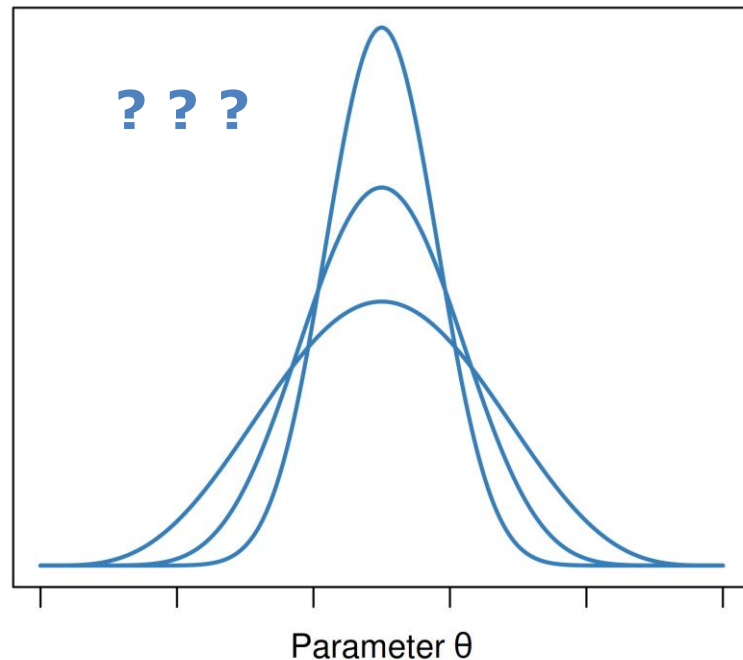
Stochastic part (residual distribution & link function)

- Ideally already provided by data type



(4) Prior distribution

- Flat, weak, or informative priors?
- You can always do better than flat priors!
- Prior predictive checks
- Often easier with scaled predictors
- brms defaults for intercept and sd often good
- Priors for effects etc. are your choice
- `default_prior(y~x+..., data=...)`

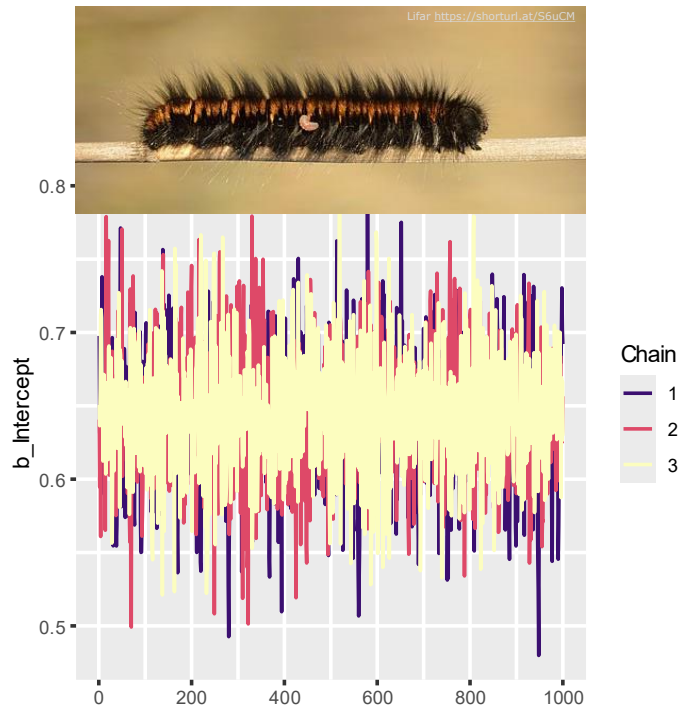


(5) Model fitting

- Run several chains
- `cores=4` will run in parallel, saves time for big models

Check convergence

- visual inspection of traceplots
- check $R_{\text{hat}} < 1.01$
- compare n_{eff} to n_{total}
 - few hundreds already good for parameter means,
but a few thousands needed for parameter quantiles
- take warning messages seriously: <https://mc-stan.org/misc/warnings.html>

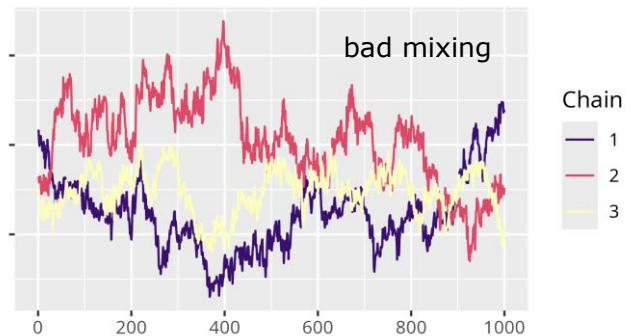
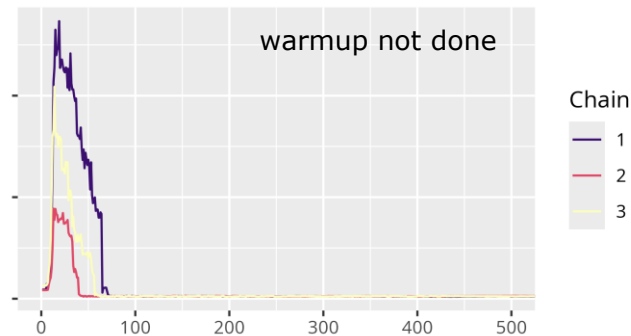


Convergence tells you if MCMC successfully approximated the posterior distribution.
Convergence does not tell you if it's a good or bad model !

(5) Model fitting

What to do if the model didn't converge?

- Provide initial values for parameters: `init=...`
- Run longer chains: `warmup=...` and `iter=...`
- Increase sampling accuracy: `control=list(adapt_delta=0.9)`
- Regularization through stricter priors



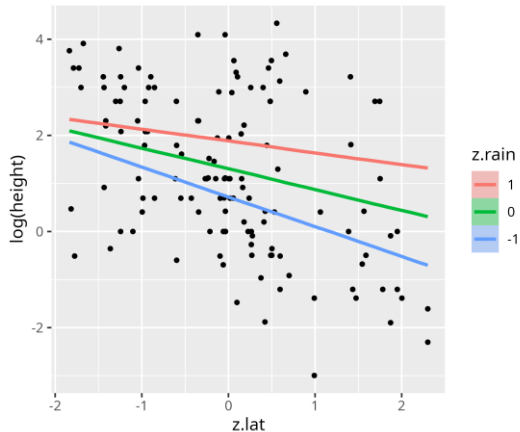
(6) Evaluate model

If, and only if MCMC converged, plot fitted / predicted against data

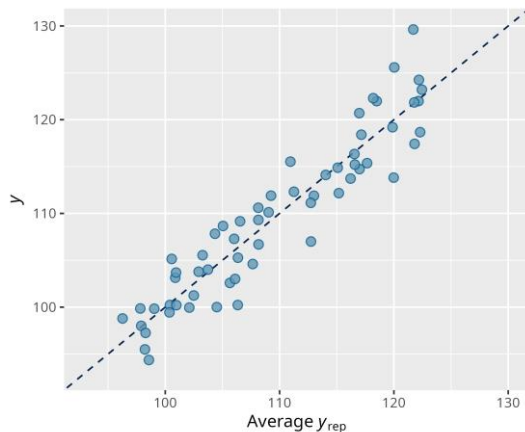
Options available from brms, performance, DHARMa packages

Bayesian statistics does **not** free you from checking model assumptions

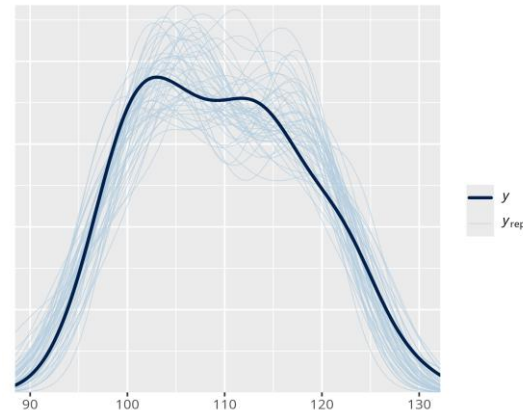
Conditional effects



Observed vs. predicted



Posterior predictive check



(6) Evaluate model

In Bayesian stats, **everything is a distribution.**

Always use posterior distribution of parameters
(not just point estimates) to compute predictions

Fitted distribution / credible intervals: deterministic model part

Predicted distribution / prediction intervals: stochastic model part



Amazing Maps
@amazingmap

Abonnieren

Due to its shape, the centre of Croatia is actually located in Bosnia and Herzegovina

[Post übersetzen](#)



(7) Quantitative statements on hypotheses

- **Statistical inference:**

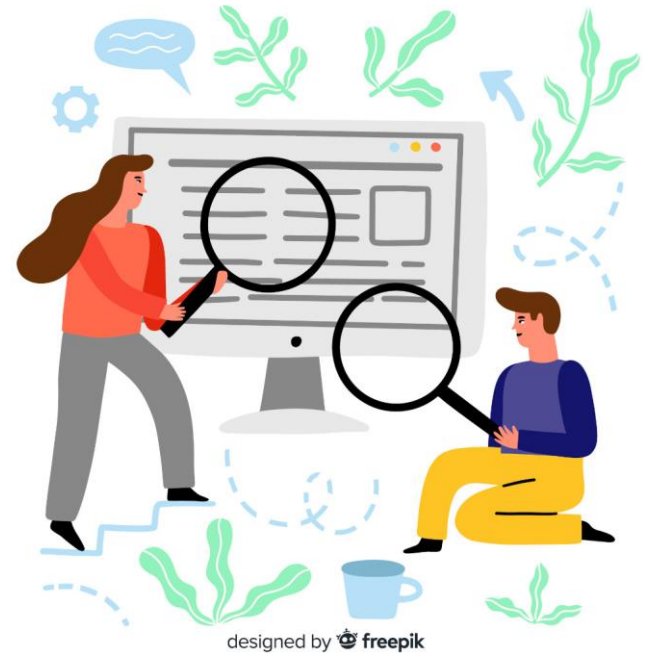
What do the data tell me about my model?

- Read $P(\theta|y)$ literally:

Quantitative statements on model (parameters θ)
and their derived quantities, given the data y .

- $P(b > 0)$, $P(\mu_1 > \mu_2)$, $P\left(\frac{\mu_1 + \mu_2}{2} > \mu_3\right)$, $P(\sigma_1 > \sigma_2)$, ...

- „Post-hoc“ analysis (emmeans) just computes and compares quantities from model parameters (level-specific means or slopes, across-level means or slopes, pairwise comparisons, group comparisons, ...)



(7) Quantitative statements on hypotheses

Model comparison with LOO

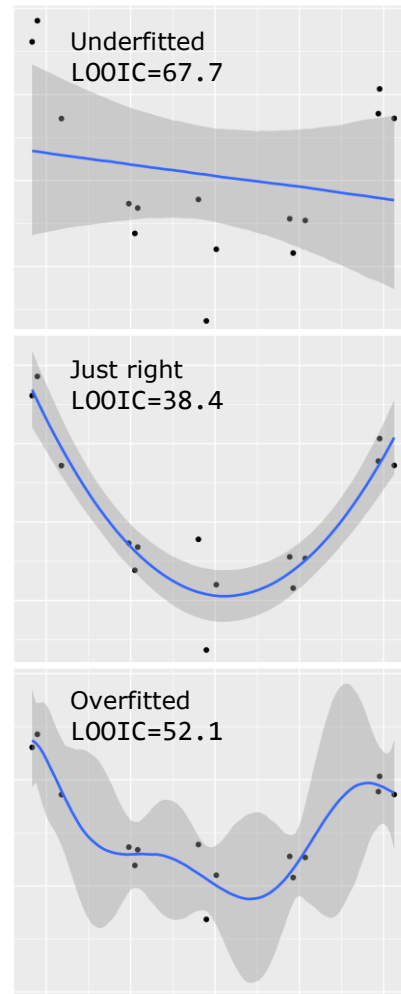
- compares expected predictive performance „elpd“ of several models
- penalizes model complexity
- LOOIC is similar to AIC (lower values = better)

Good scientific practice:

Model candidates driven by hypotheses.

Don't just fit lots of models with different predictor combinations!

„Fishing for evidence“



(8) Reporting

Methods:

We used Bayesian statistics for estimating the parameters of models (1)-(3) using the “brms” R-package. We chose normal distributions with zero mean and a standard deviation of 2 as weakly informative priors for effect sizes and brms default priors for intercept and standard deviations.

[Put priors & their justification in an SI table if you have many parameters]

We ran 4 MCMC chains of 5000 iterations with an adaptation phase of 2000 samples (12,000 posterior samples in total). We verified convergence by Gelman-Rubin statistics ($R_{hat} < 1.01$) and adequate effective sampling size (n_{eff}). For model comparison we used leave-one-out cross validation (R-package “loo”).

(8) Reporting

Results:

According to LOO, model (3) had the best predictive performance (ELPD difference to model (2): 5352.2 with standard error 709.2). We found a positive interaction effect of x1 with x2 (posterior probability $P(b_{x1:x2} > 0) = 0.970$) and a negative effect of b3 ($P(b_{x3} < 0) = 1.000$) which confirms our hypothesis X.

Parameter	mean	sd	2.5%	97.5%
Intercept	8.051	0.724	6.645	8.533
x1	0.815	0.269	0.255	1.310
x2	-0.263	0.370	-1.039	0.409
x1:x2	0.322	0.156	0.024	0.627
x3	-0.270	0.059	-0.387	-0.230

„report“ package support brms:

<https://easystats.github.io/report/reference/report.brmsfit.html>

Bayesian books

Highly recommended!

Uses his own „rethinking“ package, also based on Stan

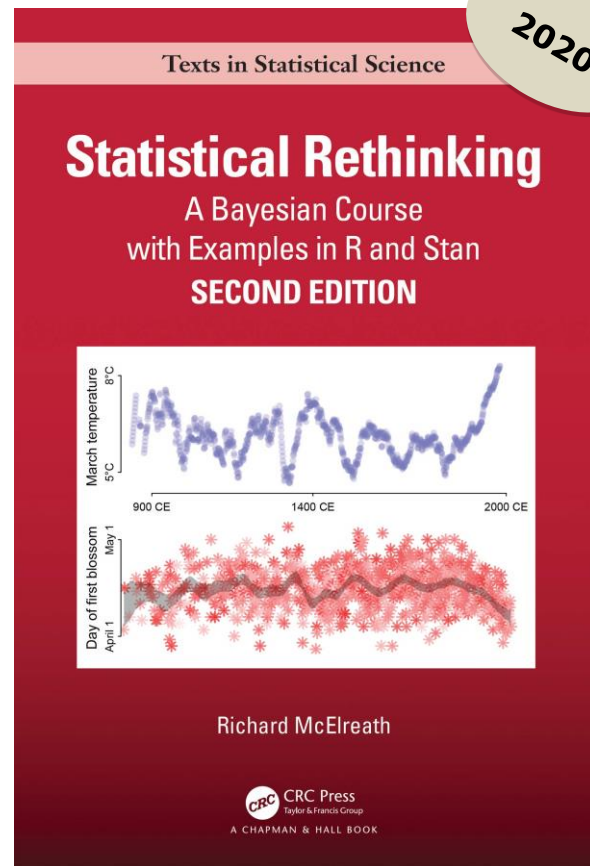
Full course & online lectures:

https://github.com/rmcelreath/stat_rethinking_2026

brms translation by Solomon Kurz:

<https://solomon.quarto.pub/sr2/>

3rd edition available soon?

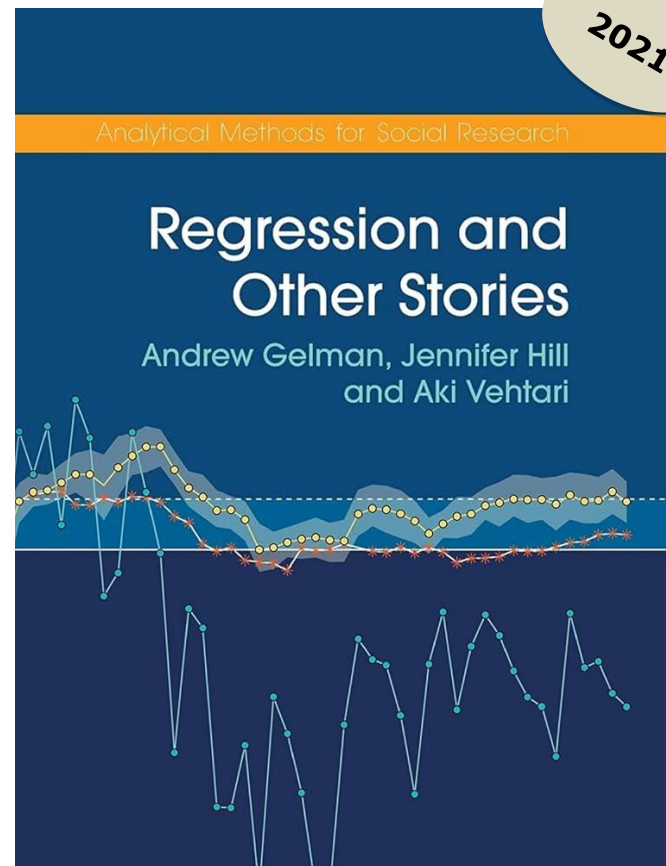


Bayesian books

Examples more focused on social & political sciences,
but still general enough

Uses „rstanarm“ package

Update of their other book on mixed-effects models
„Advanced Regression and Multilevel Models“ will
come out this year

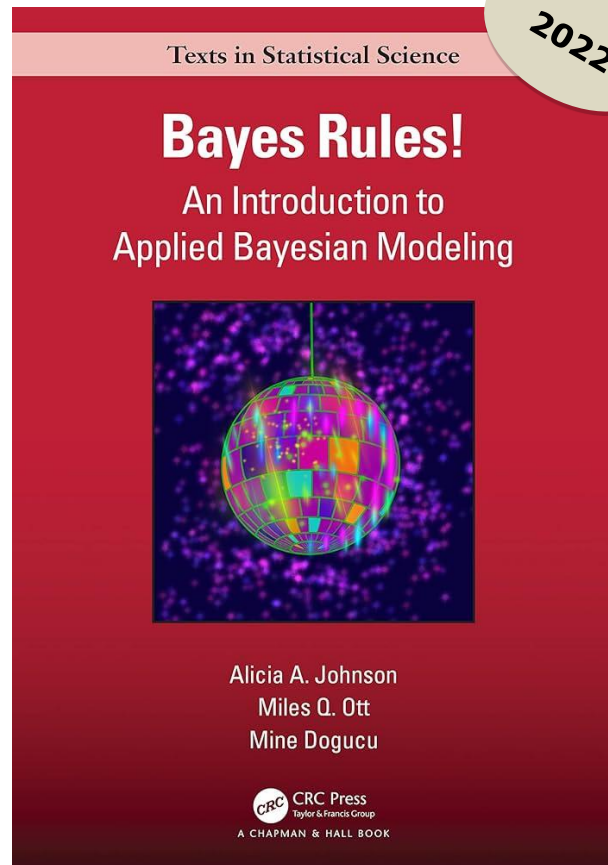


Bayesian books

Free online version available:

<https://www.bayesrulesbook.com/>

Uses „rstanarm“ and Stan



Bayesian books

Free online version available:

<https://paulbuerkner.com/software/brms-book/>

Work-in-progress, full version available next year

2026

**The brms Book: Applied Bayesian Regression
Modelling Using R and Stan (Early Draft)**

Paul-Christian Bürkner

2024-10-30

Ecology books

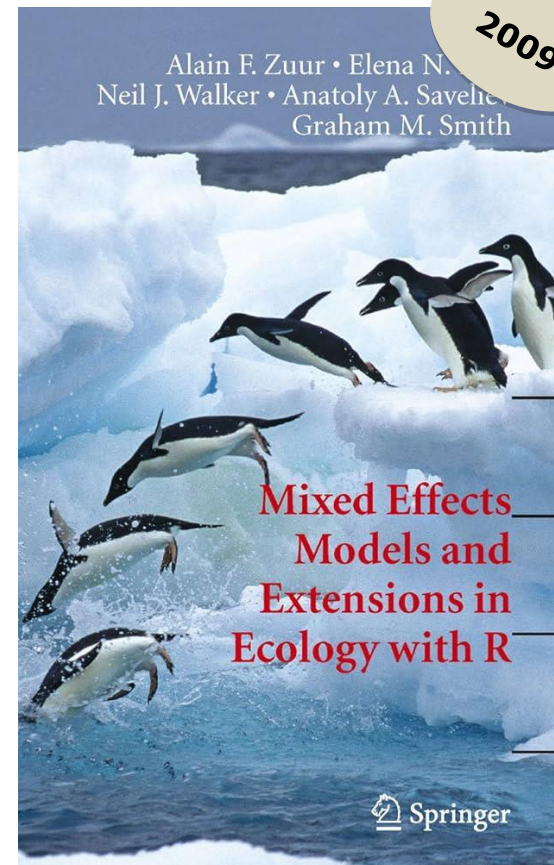
Purely frequentist

Still a classic for specific ecological problems
(mixed effects, autocorrelation, zero-inflation, ...)

Many best-practice examples with step-by-step guide
of data analysis and statistical inference

Data package „AED“ discontinued, but use
> `remotes::install_github("romunov/AED")`

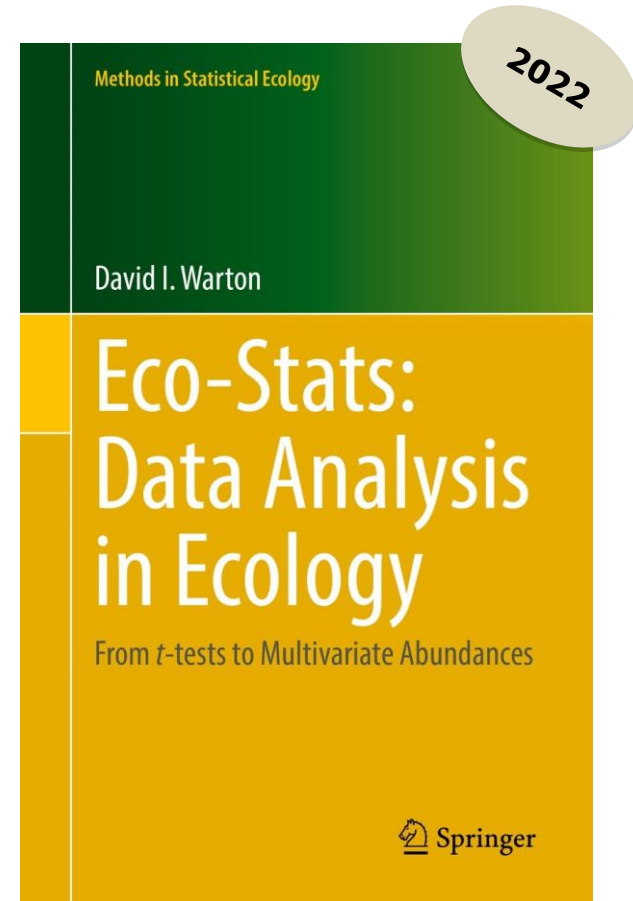
→ Try to replicate some analyses with brms



Ecology books

Purely frequentist

Half of the book dedicated to multivariate analysis!

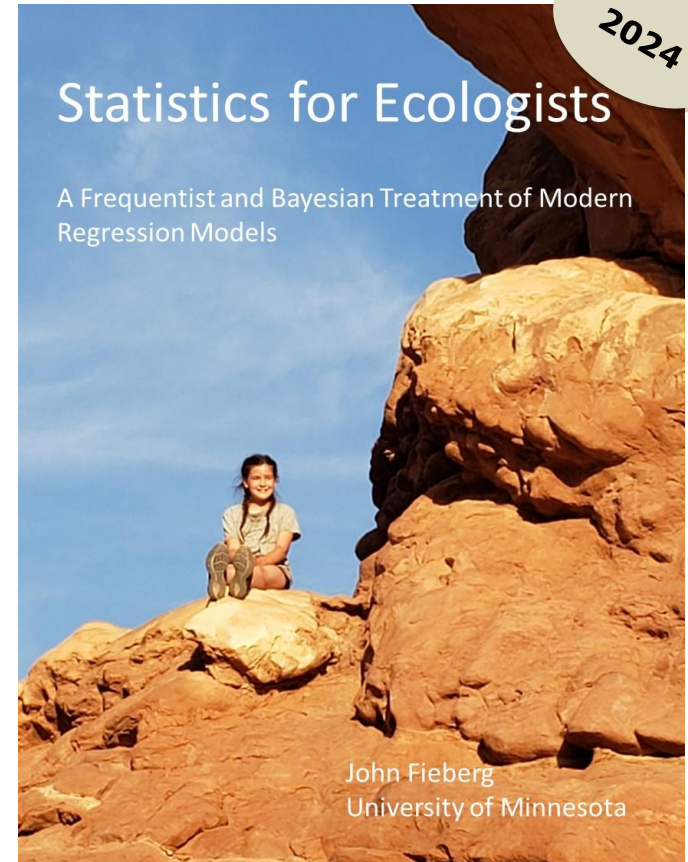


Ecology books

Free online version available:

<https://statistics4ecologists-v3.netlify.app/>

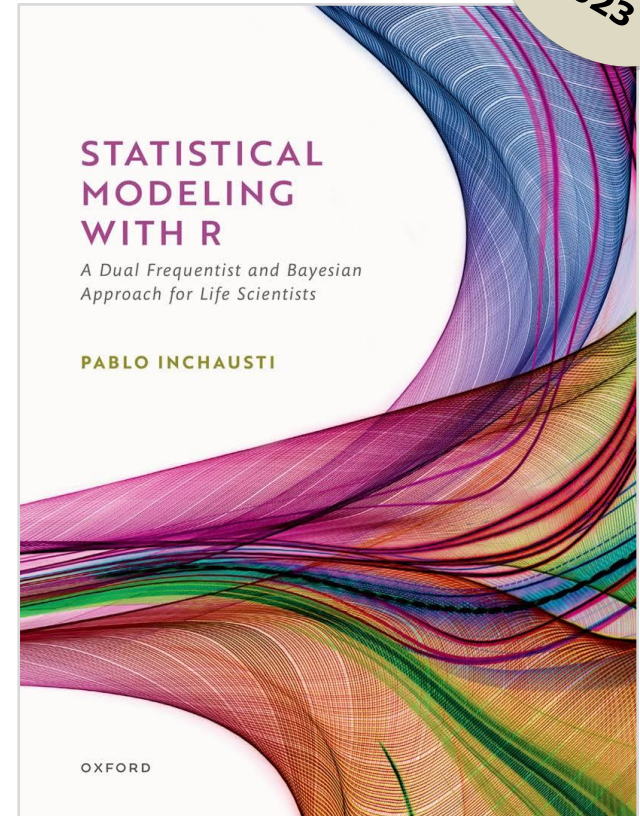
Mostly frequentist, just a brief Bayesian chapter



A Bayesian ecology-related book

A dual frequentist and Bayesian approach

Uses brms !

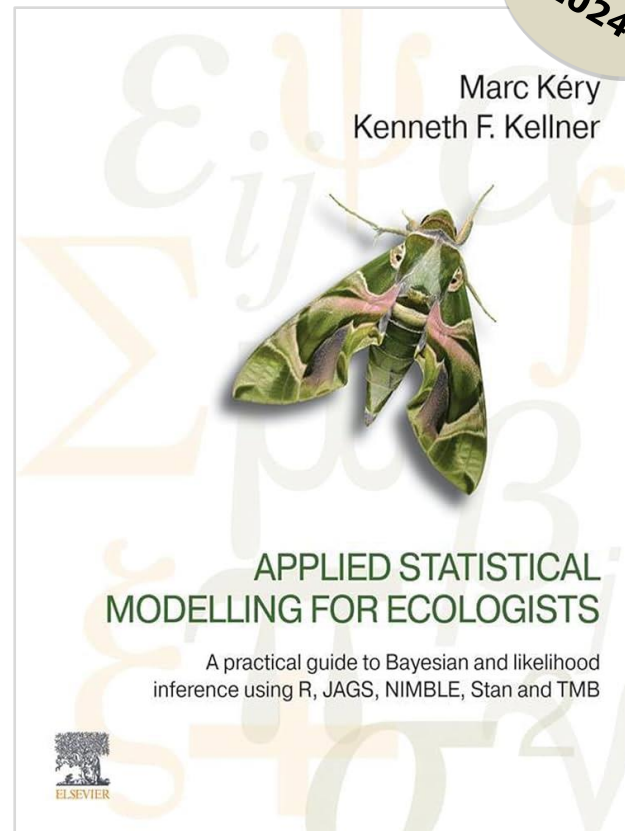


A Bayesian ecology book

Successor of Kery's 2010 WinBUGS book

Uses Stan, NIMBLE, etc, but no brms

If you're looking for an accessible & concise introduction to coding ecologically relevant models, this probably is the one.



Online Resources

- Stan forums <https://discourse.mc-stan.org/>
- brms section <https://discourse.mc-stan.org/c/interfaces/brms>
- some tutorials: <https://mc-stan.org/learn-stan/tutorials.html>



Interfaces ▾

brms ▾

tags ▾

Latest

New (1)

Unread (3)

Hot

Topic

📌 About the brms category

This category is for questions regarding the installation and use of brms. If you have installation issues then please provide as much information about your system as possible.

Consulting

Got stuck with your data / model / analysis ?

→ **Contact me!** benjamin.rosenbaum@idiv.de

I offer statistical consulting for all
iDiv-affiliated researchers
(Bayesian & Frequentist)



Source: freepik.com

Take home:

Frequentist stats are often like:

Test *ABC* can do *XYZ*.

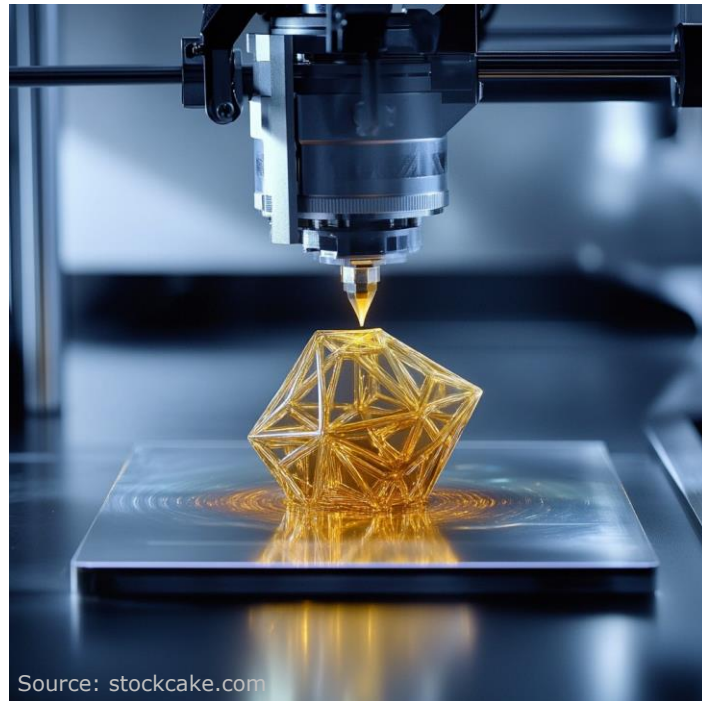
Bring hypotheses and data in alignment with
test's assumptions

Bayesian (& other!) stats should be like:

Model the data-generating process.

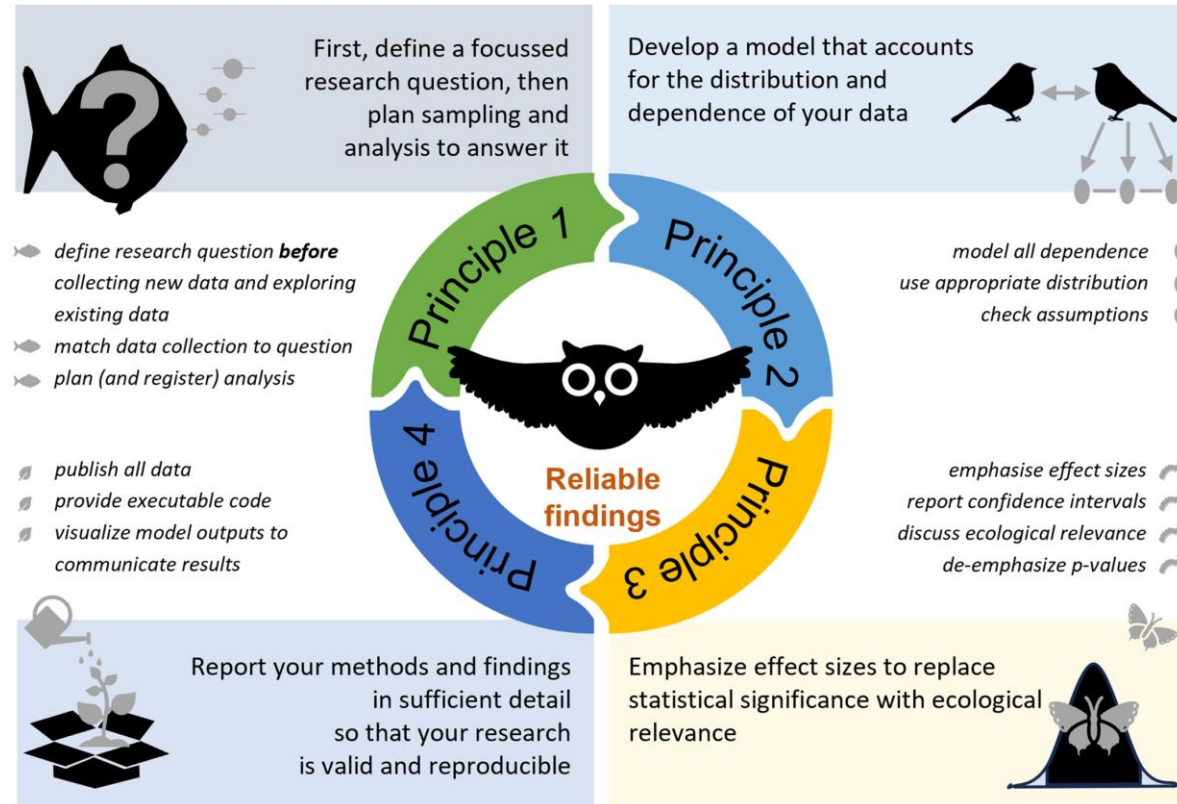
Derive something quantifiable (parameters or predictions)
from the model that describes your original research question.

The Bayesian 3D printer



Source: stockcake.com

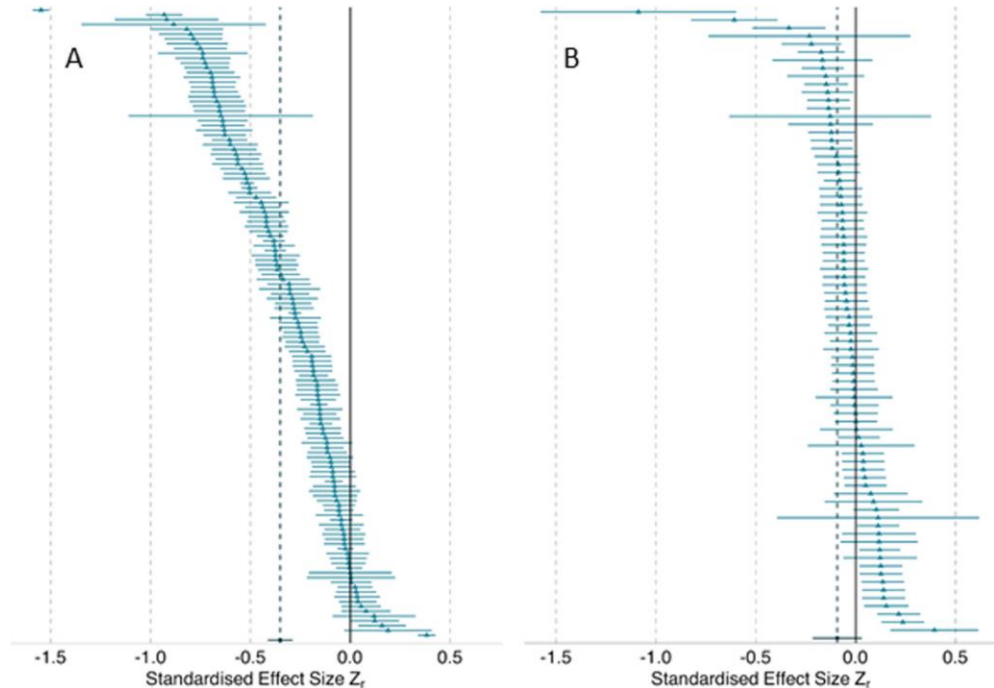
Better scientific practice





Same data, different analysts: variation in effect sizes due to analytical decisions in ecology and evolutionary biology

Gould et al. (2025) BMC Biology
<https://doi.org/10.1186/s12915-024-02101-x>



There are many ways to to mess up an analysis, but there is **no single right way** to do it, either.

Good luck !

